

Android Project Specification: Quick Math

Objective: Build a complete Android Application in Kotlin using Jetpack Compose that implements a "forced learning" mechanism through math quiz overlays.

1. Core Identity & Compliance

- **App Name:** Quick Math
- **Privacy:** Fully offline after setup. Zero personal data collected/shared.
- **Age Gate:** 18+ only.
- **UI Style:** Strict WhatsApp Dark Mode (Primary: #0b141b, Secondary: #202c33, Accent: #00a884, Text: #e9edef).

2. Localization (Trilingual)

Implement a locale-switching mechanism for:

- **English**
- **Hindi**
- **Telugu**
- *Requirement:* On first launch, show a "Select Language" screen before any other setup.

3. Permissions & Background Logic

Implement the following permission flows with custom user-friendly dialogs:

- **System Overlay Permission:** Required for math cards to pop up over other apps.
- **Ignore Battery Optimizations:** Required for the "Strict Mode" and "Background Run."
- **Quick Settings Tile:** Code to register a TileService so the user can start/stop sessions from the notification tray.

4. Onboarding & Security

- **OTP Verification:** A screen to verify the user via phone number. (Mock the backend API but provide the UI and logic for SMS verification intent).
- **Setup Flow:** Language Selection -> 18+ Confirmation -> Phone OTP -> Permissions Granting (Overlay & Background).

5. Home Screen (WhatsApp UI Style)

- **Header:** "Quick Math" with a Menu icon (Settings, Share, Contact Us, Feedback, About Us).
- **Session Controls:**
 - Large "START SESSION" button.

- **Dropdown Menus:**
 - "Session Duration": Options (5, 10, 15 min).
 - "One Card Every": Options (30s, 1m, 2m, 5m).
 - "Answer Time": Options (5s, 10s, 30s, 60s).
- **Status Indicator:** Show if the service is currently running.

6. The "Math Quiz" Overlay Logic

- When a trigger hits (based on "One Card Every" timer):
 - **Vibration:** Trigger maximum intensity vibration for the duration of the card.
 - **Audio:** Play a TTS (Text-to-Speech) or audio clip: "Please answer quickly."
 - **Visuals (Strict vs. Fun):**
 - **Strict Mode:** High-contrast math problem, full-screen or half-screen based on settings.
 - **Fun Popups Mode:** Two variants:
 1. Black background with bouncing Red and Green bubbles (Phone call style).
 2. White smoking/buffering ring animation around the math problem.
- **Math Logic:** Generate random arithmetic (Addition, Subtraction, Multiplication) suitable for quick solving.

7. Settings & Features

- **Mode Toggle:** "Strict Mode" vs "Fun Popups Mode".
 - **Strict Mode Trial:** Logic for a 4-day free trial (no credit card required). Enable Quick Tile on activation.
- **Display Settings:** Toggle between Half Screen (50% height) and Full Screen (90% height).
- **Audio/Vibration:** Master toggles for sound and haptics.
- **Language:** Option to change the app language.
- **Menu Functionalities:**
 - **Share:** Intent to share app link.
 - **Contact/Feedback:** Simple form or Email intent.
 - **About:** Version info and "Privacy First" disclaimer.

8. Technical Stack Requirements

- **Language:** Kotlin
- **UI:** Jetpack Compose (Material 3)
- **Concurrency:** Coroutines for timers.
- **Service:** A ForegroundService to keep the app alive during sessions.
- **Data:** DataStore or SharedPreferences for local settings.

Instructions for Gemini CLI: Please generate the complete project structure, including the AndroidManifest.xml, build.gradle.kts, and all necessary Kotlin files (ViewModels, Services, Compose UI) to fulfill these requirements.